

3-dimensional arrays:

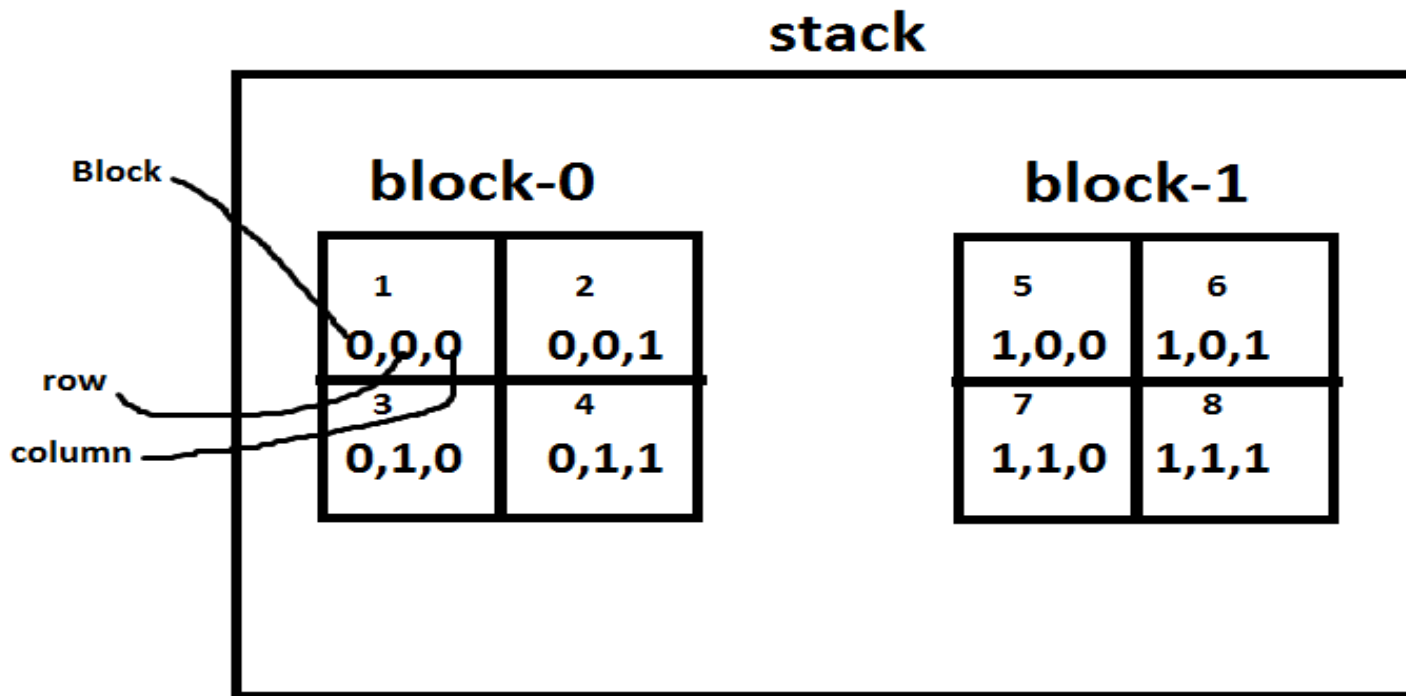
An array with several blocks, rows and columns.

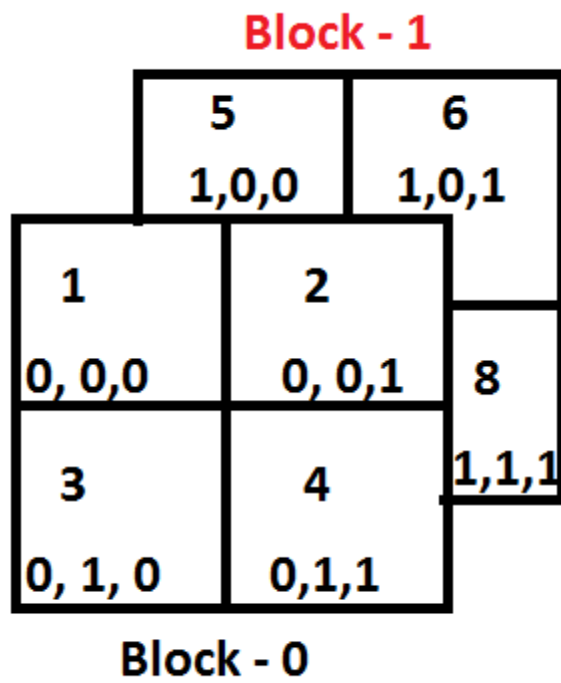
An array with 3 subscripting operators **[] [] []**.

Syntax:

datatype variable [blocks] [rows] [columns];

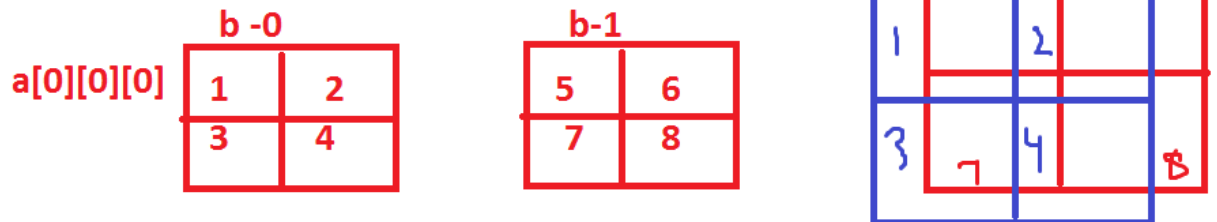
Eg: int a[2][2][2]={1,2,3,4,5,6,7,8};





Eg:

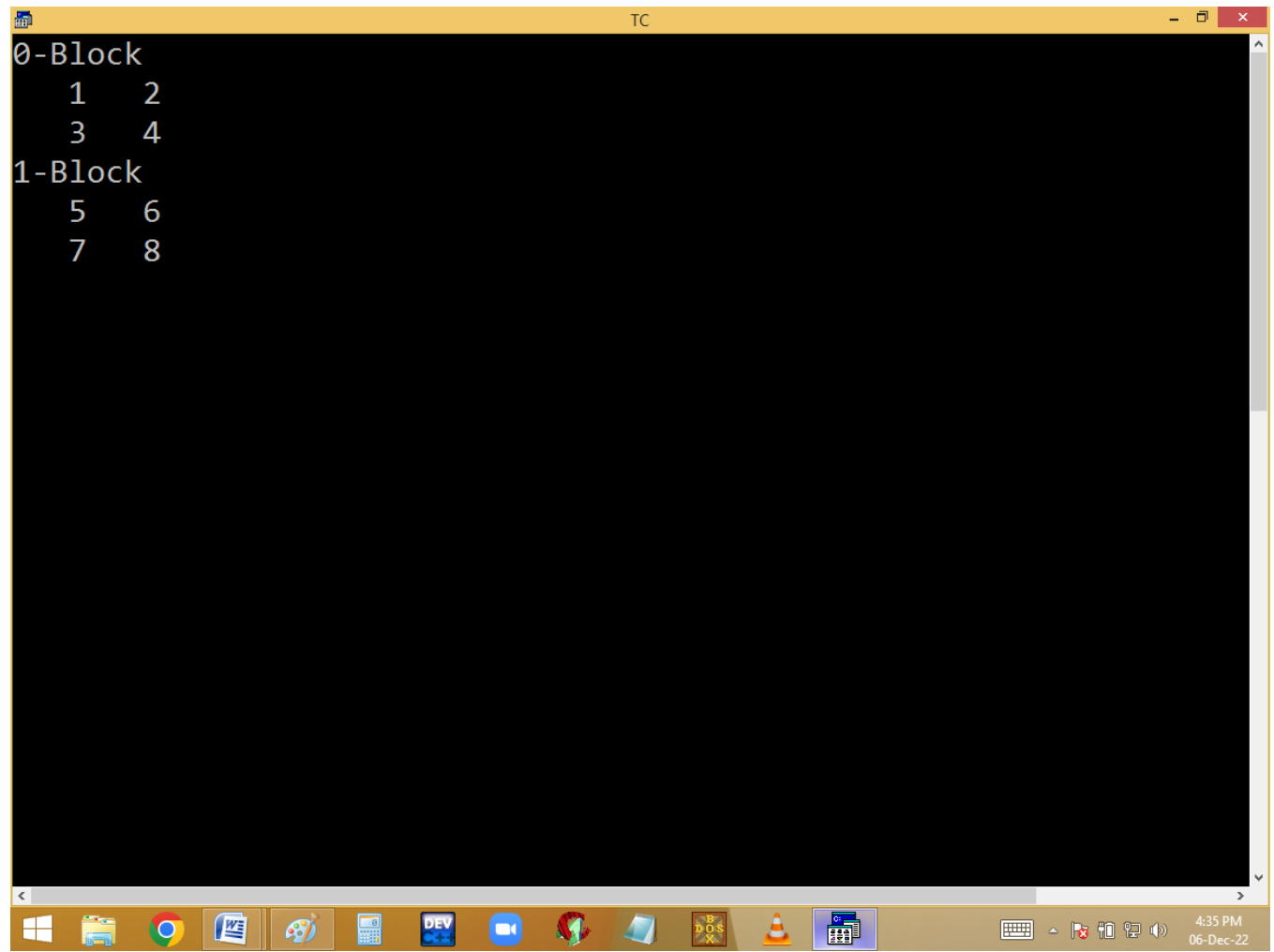
rows
 int a[2][2][2]; an array with 3 subscripts [].
 blocks columns



eg: int class[2][60][6];
 datatype class[sections][stus][marks];

```

TC
File Edit Run Compile Project Options Debug Break/
Edit
Line 1 Col 2 Insert Indent Tab Fill Unindent * E:NONAM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][2][2]={1,2,3,4,5,6,7,8},b,r,c;
clrscr();
for(b=0;b<2;b++)
{
printf("%d-Block\n",b);
for(r=0;r<2;r++)
{
for(c=0;c<2;c++) printf("%4d",a[b][r][c]);
printf("\n");
}
}
getch();
}
Watch
  
```



The screenshot shows a Windows 10 desktop with a taskbar at the bottom. A terminal window titled 'TC' is open, displaying a 4D array structure. The array is organized into two main blocks: '0-Block' and '1-Block'. Each block contains two rows of two numbers each. The '0-Block' contains the numbers 1, 2, 3, and 4. The '1-Block' contains the numbers 5, 6, 7, and 8. The terminal window has a yellow title bar and a black background. The taskbar at the bottom shows various application icons, including the Start button, File Explorer, Google Chrome, Microsoft Word, Paint, Calculator, DEV, a video call icon, a game icon, a folder icon, a game icon, a VLC media player icon, and a calendar icon. The system clock in the bottom right corner shows the time as 4:35 PM on 06-Dec-22.

```
TC
0-Block
  1  2
  3  4
1-Block
  5  6
  7  8
```

4-dimensional array:

An array with several sets, blocks, rows and columns.

An array with 4 subscripting operators **[] [] [] []**

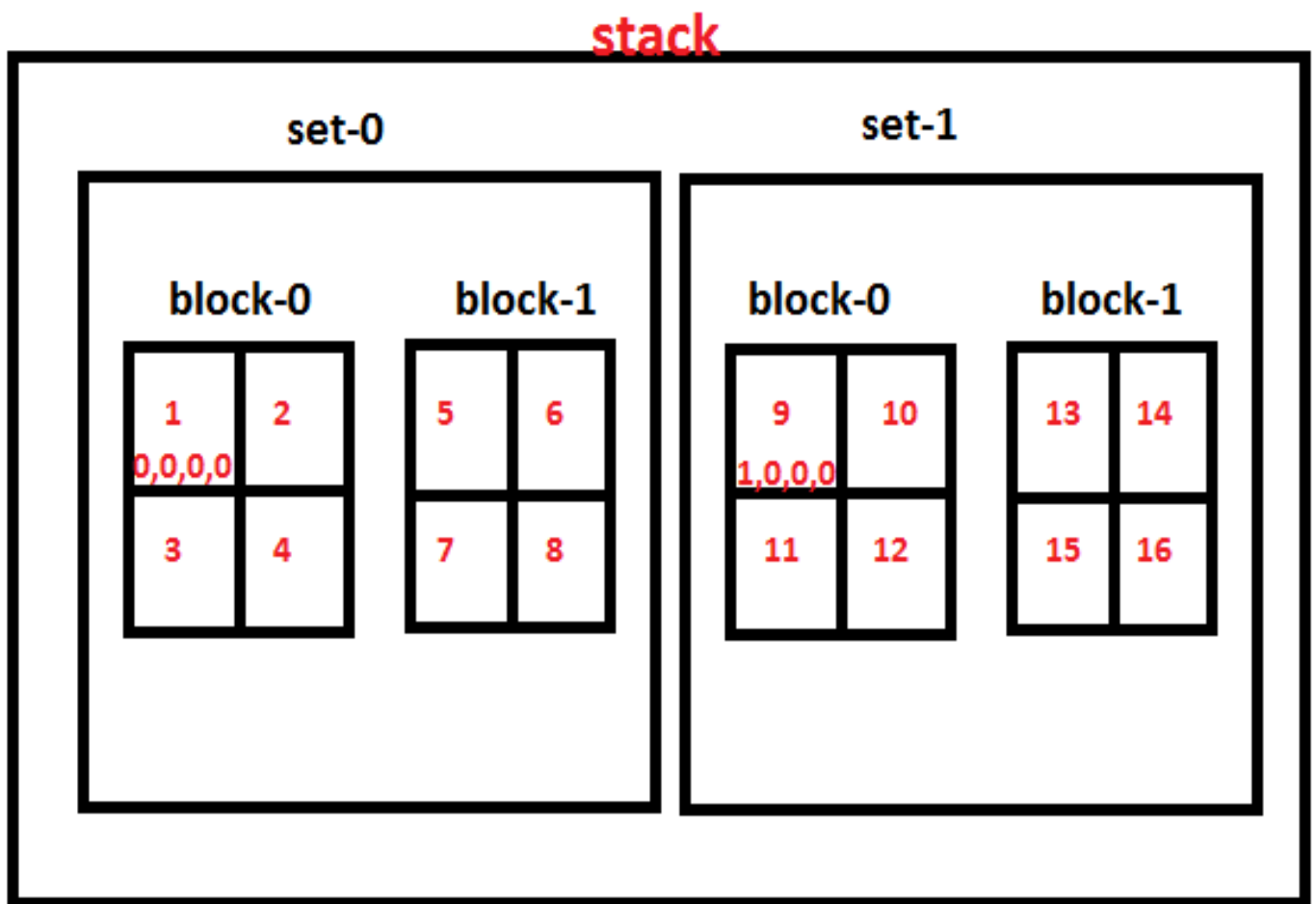
▪

Syntax:

datatype variable [sets] [blocks] [rows] [cols];

eg:

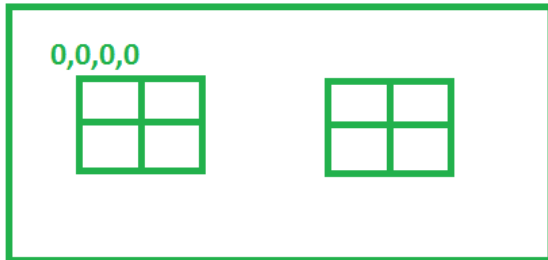
int a[2] [2] [2] [2]= {1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16};



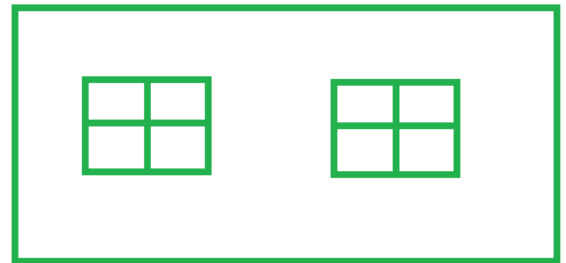
```
datatype var[class][sec][stu][marks];  
int      school[5][2][60][6];
```

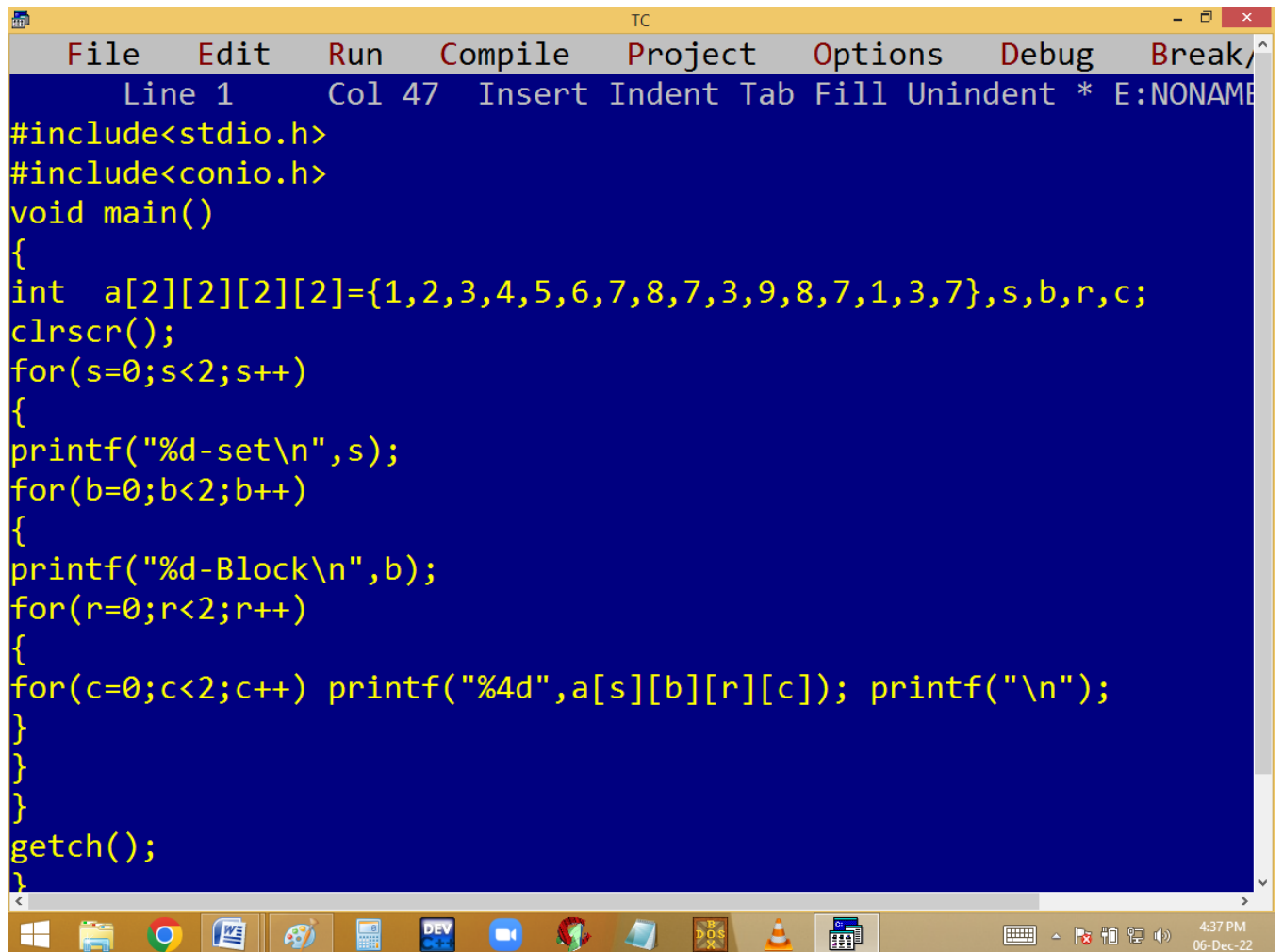
set
rows
int a[2][2][2][2]; an array with 3 subscripts [].
blocks columns

set-0



set-1

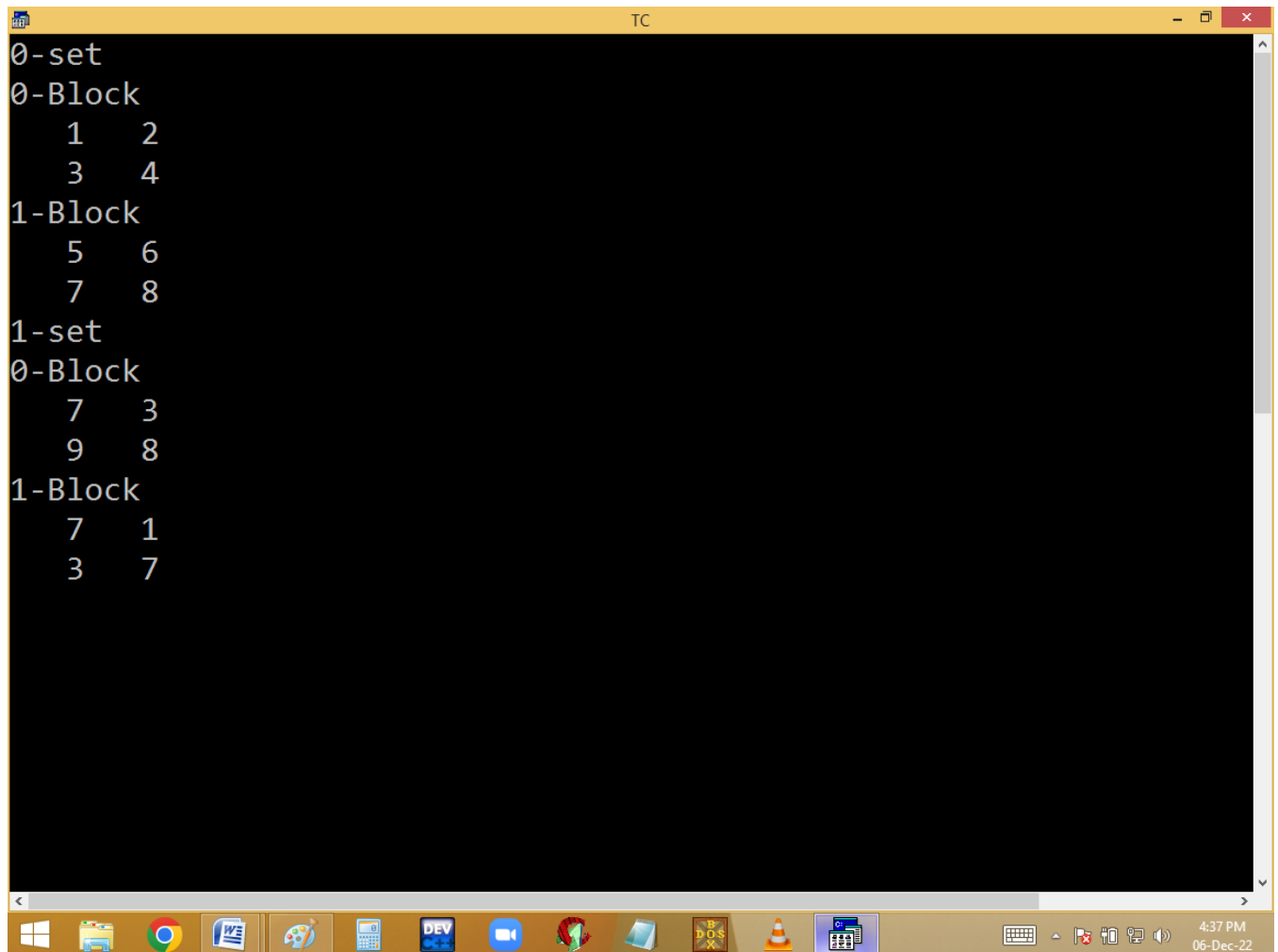




The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/". The status bar at the top indicates "Line 1", "Col 47", and "Insert Indent Tab Fill Unindent * E:NONAME". The main editing area has a dark blue background with yellow text. The code is a C program that defines a 2D array 'a' and iterates through it using nested loops. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int  a[2][2][2][2]={1,2,3,4,5,6,7,8,7,3,9,8,7,1,3,7},s,b,r,c;
clrscr();
for(s=0;s<2;s++)
{
printf("%d-set\n",s);
for(b=0;b<2;b++)
{
printf("%d-Block\n",b);
for(r=0;r<2;r++)
{
for(c=0;c<2;c++) printf("%4d",a[s][b][r][c]); printf("\n");
}
}
}
getch();
}
```

The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 4:37 PM on 06-Dec-22.



```
0-set
0-Block
    1    2
    3    4
1-Block
    5    6
    7    8
1-set
0-Block
    7    3
    9    8
1-Block
    7    1
    3    7
```

The screenshot shows a Turbo C++ (TC) IDE window with a black background and white text. The code is as follows:

```
0-set
0-Block
    1    2
    3    4
1-Block
    5    6
    7    8
1-set
0-Block
    7    3
    9    8
1-Block
    7    1
    3    7
```

The window title bar says "TC". The taskbar at the bottom shows various icons including Windows, File Explorer, Chrome, Word, Paint, Calculator, DEV, a blue icon, a red icon, a folder icon, a DOS icon, a VLC icon, and a calendar icon. The system clock in the bottom right corner shows "4:37 PM 06-Dec-22".

Finding n students grade

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
void tummy(float a){float *p=&a;}
```

```
void main()
```



```
{  
float stu[10][10]={0};  
char name[10][30];  
int n,i,p,j;  
clrscr();  
printf("Enter no of students ");scanf("%d",&n);  
for(i=0;i<n;i++)  
{  
printf("Enter %d stu id, name, 6 sub marks ", i+1);  
scanf("%f%s",&stu[i][0],name[i]);  
for(j=1;j<=6;j++)  
{scanf("%f",&stu[i][j]);  
stu[i][7]+=stu[i][j];  
if(stu[i][j]<35)stu[i][9]=-1;  
}  
stu[i][8]=stu[i][7]/6.0;  
}
```

```
puts("Id\tName\tTotal\tAvg\tPass/Fail");  
puts("-----");  
for(i=0;i<n;i++)  
{  
    printf("%.0f\t%s\t%.0f\t%.2f\t%s\n",stu[i][0],name[i],stu[i][7],stu[i][8],  
    stu[i][9]==0?"Pass":"Fail");  
}  
puts("-----Thank you-----");  
getch();  
}
```

```
TC
Enter no of students 3
Enter 1 stu id, name, 6 sub marks 1 ajay 88 99 90 99 88 99
Enter 2 stu id, name, 6 sub marks 2 vijay 22 33 43 22 33 22
Enter 3 stu id, name, 6 sub marks 3 neha 54 45 65 56 66 77
Id      Name      Total      Avg      Pass/Fail
-----
1      ajay      563      93.83      Pass
2      vijay      175      29.17      Fail
3      neha      363      60.50      Pass
-----
Thank you-----
```

Activate Windows
Go to PC settings to activate Windows.

STRINGS

- A group of characters is called string.
- It is one dimensional character array.
- It is alpha-numeric.
- It is an implicit pointer.
- It is a derived data type.

Note:

- One byte should be left for Null char(**\0**). Otherwise we are getting garbage or junk values. Null char indicates string is completed.
- **String variable Size can't be less than string. Otherwise we are getting error.**
- Using **=** operator, we can't copy a string into another. We have to use strcpy() or copy character by character manually.
- Using **==** (comparison) operator, we can't compare two strings. Use strcmp() or compare the characters one by one manually.

Syntax:

char variable [size] = "string";

or

char variable[]="string";

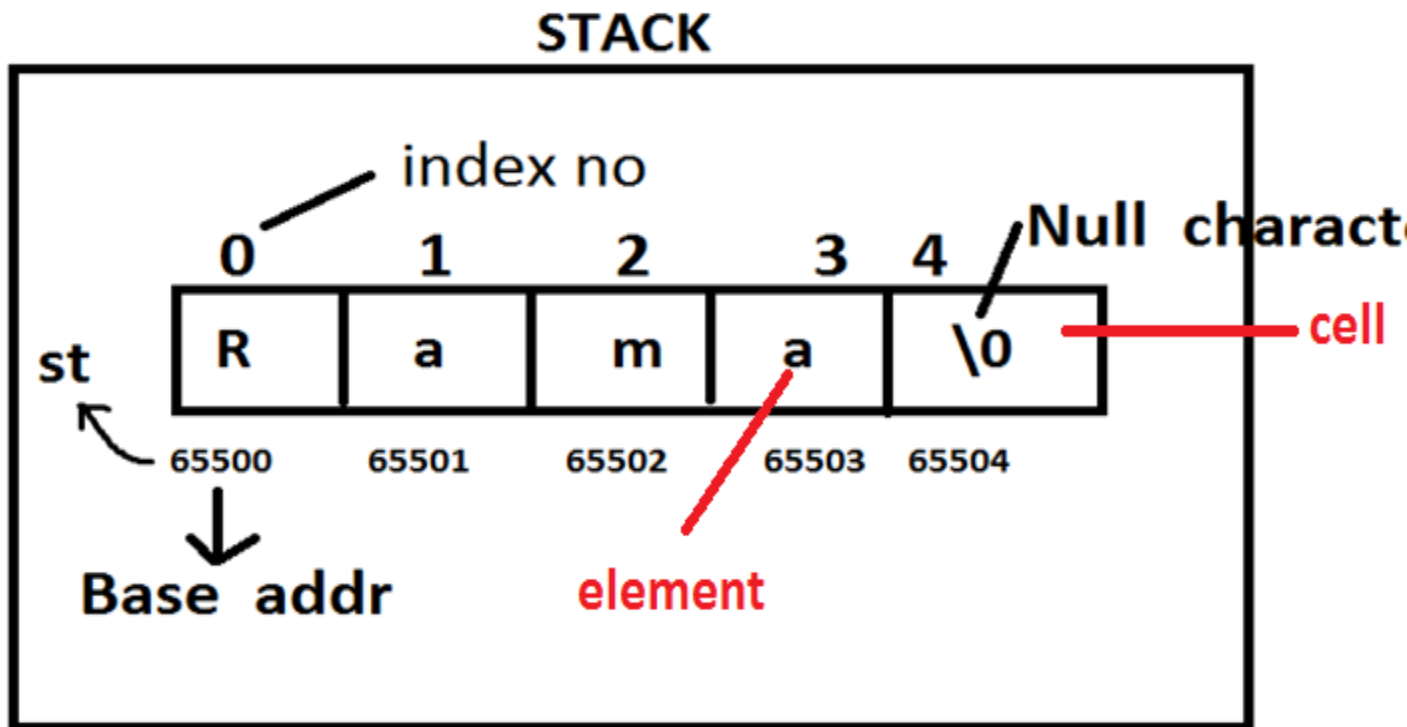
Eg: char st[5] = "Rama";

Why?

In C, a string is an array of characters stored in memory.

When you use == to compare two strings, it compares their memory addresses, not their actual contents.

Since the two strings are stored at different memory locations, str1 == str2 will return false, even if the contents are the same.



Note: String is implicit pointer because of string variable stores base address.

String declaration methods:

`char st[5] = "rama";` Ok

`char st[20] = "Naresh It";` Ok

`char st[4] = { 'r', 'a', 'm' };` Ok → char array.

char st[3]= "ram"; It gives garbage values in printing.

char st [3] = "rama"; error

char st[0]; error

char st[0]="abc"; Ok

char st[-5]; error

char st[5.5]; error

char st[5%3]; Ok → char st[2];

char st[3+2]; → st[5] → Ok

char st[] ="Ram"; Ok.

char st[] ; error

int n=20;

char st[n]; No

#define n 20

char st[n]; Ok

Note: String variable size always constant positive integer value.

in turbo if the size is even then it is properly show the output, but if you enter odd number then output with garbage value print